

Experiment No-10(a)

Verification of Linearity of a Discrete System

AIM: Verify the Linearity of a given Discrete System.

Software Required:

Mat lab software 7.0 and above

Theory:

LINEARITY PROPERTY:

Any system is said to be linear if it satisfies the superposition principal. superposition principal state that Response to a weighted sum of input signal equal to the corresponding weighted sum of the outputs of the system to each of the individual input signals.

If $x(n)$ is a input signal and $y(n)$ is a output signal then

$$y(n)=T[x(n)]$$

$$y_1(n)=T[x_1(n)] \text{ and } y_2(n)=T[x_2(n)]$$

$$x_3=[a*x_1(n) +b *x_2(n)]$$

$$Y_3(n)= T [x_3(n)]$$

$$T [a*x_1(n)+b*x_2(n)] = a y_1(n)+ b y_2(n)$$

Program:

% Verification of Linearity of a given System.

% a) $y(n)=nx(n)$ b) $y=x^2(n)$

clc;

clear all;

close all;

n=0:40;

a1=input('enter the scaling factor a1=');

a2=input('enter the scaling factor a2=');

x1=cos(2*pi*0.1*n);

x2=cos(2*pi*0.4*n);

x3=a1*x1+a2*x2;

%y(n)=n.x(n);

y1=n.*x1;

y2=n.*x2;

y3=n.*x3;

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yt=a1*y1+a2*y2;
yt=round(yt);
y3=round(y3);
if y3==yt
disp('given system [y(n)=n.x(n)]is Linear');
else
disp('given system [y(n)=n.x(n)]is non Linear');
end
%y(n)=x(n).^2
x1=[1 2 3 4 5];
x2=[1 4 7 6 4];

x3=a1*x1+a2*x2;
y1=x1.^2;
y2=x2.^2;
y3=x3.^2;
yt=a1*y1+a2*y2;
if y3==yt
disp('given system [y(n)=x(n).^2 ]is Linear');
else
disp('given system is [y(n)=x(n).^2 ]non Linear');
end

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Result: The Linearity of a given Discrete System is verified.

Output:

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enter the scaling factor a1=3
enter the scaling factor a2=5
given system [y(n)=n.x(n)]is Linear
given system is [y(n)=x(n).^2 ]non Linear

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VIVA QUESTIONS:-

1. Define linear system with example?
 2. Define non- linear system with example?
 3. Define super position principle?
 4. Give mathematical expression for Linearity.
- Identity the system $y(n)=[x(n)]^3$ is linear or non-linear system